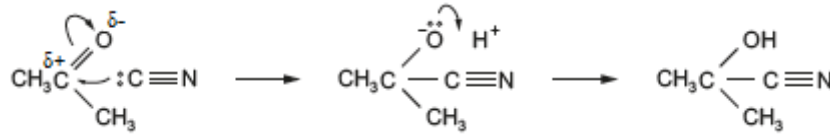
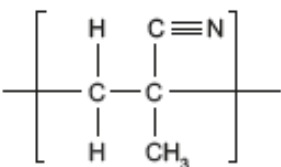


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11.	(a)	(i)		 nucleophilic addition (1) curly arrows (1) partial / full charges (1)	3			3		
		(ii)		incomplete extraction by ethoxyethane (1) incomplete distillation / decomposition of product (1)			2	2		2
		(iii)		moles of propanone used = $17.4 / 58.06 = 0.300$ (1) 1:1 molar ratio $\Rightarrow$ 0.300 mol of 2-hydroxy-2-methylpropanenitrile mass of 2-hydroxy-2-methylpropanenitrile = $0.300 \times 85.07 = 25.5$ g (1) percentage yield = $18.6 \times 100 / 25.5 = 73 \%$ (1)		3		3	2	
		(iv)				1		1		

Question				Marking details	Marks available						
					AO1	AO2	AO3	Total	Maths	Prac	
	(b)			OH CH₃CH₂C H — CH₂—C O OH (1) balancing i.e. n monomer units and nH₂O molecules on product side (1) <td></td> <td></td> <td></td> <td>1</td> <td>2</td> <td></td> <td></td>				1	2		
	(c)			in condensation polymerisation a small molecule e.g. water is lost; this does not occur with addition polymerisation accept alternatives e.g. different numbers of monomer types, functional groups, atom economy must refer to both types of polymerisation	1			1			
	(d)	(i)		surround the flask with a cold water bath	1			1		1	
		(ii)		1 mol $\equiv$ 24500 cm³ of hydrogen from 2 mol of the alcohol 184 cm³ from $184 \times 2 / 24500 = 0.015$ mol of the alcohol (1) $M_r = 0.900 / 0.015 = 60$ (1)							
		(iii)		H₃C H C H₃C OH ecf possible from incorrect M_r in part (ii) <td></td><td></td><td></td><td>1</td><td>1</td><td></td><td>1</td>				1	1		1
				Question 11 total	5	8	3	16	3	6	